

EDUCATION

Carnegie Mellon University <i>M.S. in Software Engineering GPA: 4.0 / 4.0</i> Selected Coursework: Large Language Model Systems, Distributed Systems, Machine Learning	Expected May 2027
University of Wisconsin–Madison <i>B.S. in Computer Science & B.S. in Computer Engineering GPA: 3.72 / 4.0</i> Selected Coursework: Algorithms, Data Structure & Programming III, Large Language Models in Practice	May 2024

WORK EXPERIENCE

NewsBreak <i>Backend Engineer Intern</i>	May – August 2026 <i>Mountain View, CA</i>
- Built deterministic span-based entity resolution over a 206-entity catalog as the search query rewriter, taking real logged queries that previously returned nothing from 0% to 100% P@1 at 0.4 ms; an eval harness on the real catalog scored 28/28 recall after measurements ruled out embeddings. - Shipped agentic LLM search on one SSE endpoint: deterministic RAG over hybrid retrieval (keyword, full-text, and bge vectors fused with RRF) that degrades to surviving legs, a bounded tool-use agent for follow-ups, and server-resolved citations so the model cannot author a URL. - Built an LLM pipeline that labels every incoming article (happened / speculative / no_event plus 0–100 importance; ~1,200 LLM calls/day) and logs why each item fired or not; calibrated it on 4,100+ decision rows in production shadow and shipped it live, cutting filler 34% → 4%. - Built the FastAPI / MongoDB backend from its first commit to 89 endpoints on AWS EKS with ~1,450 offline tests; re-keyed a per-user cache to a shared one, 78× smaller, as pod memory neared 512 MiB. - Diagnosed a 65% failure rate in another team’s LLM pipeline, via Langfuse traces, as an unbounded reasoning loop (p95 1,197 tokens vs. runaways at 16,384); fixed a 5-day silent CronJob outage with paired job and scheduler deadlines. - Sole SwiftUI engineer for SportsBreak (~6,000 installs, peak DAU 1,119) and primary backend author; 344 PRs (340 merged) across five repos; its LLM morning digest reached 5,026 users.	
Haddee Education <i>Engineering Lead</i>	June – September 2025 <i>San Diego, CA</i>
Led an 8-intern team to ship a React, Next.js, FastAPI, and Supabase web app from 0→1 on GCP App Engine; designed the schemas and REST APIs for its LangChain resume parsing and AI job matching.	
ZenAI — MiraclePlus (Formerly YC China) 2023 Fall Batch <i>Full Stack Developer</i>	September – December 2023 <i>Beijing, China</i>
Built a VC-backed startup’s LLM mock-interview web app from 0→1 as the sole engineer (React, Next.js, Supabase); iterated AI scoring with recruiters until three 200–500-person companies signed.	

RESEARCH EXPERIENCE

Control & Learning Group — Carnegie Mellon University <i>Graduate Researcher Advisor: Prof. Yorie Nakahira</i>	February – June 2026
Mapped natural-language instructions to formal safety specs with multi-turn LLMs and implemented Bayesian estimators for real-time road-friction adaptation (LuGre tire models).	

SKILLS

Languages	Python, Java, TypeScript, JavaScript, Go, SQL, Swift, C
AI & LLMs	RAG, hybrid retrieval, LLM tool calling / agents, bge embeddings, Langfuse, LangChain
Backend & Data	FastAPI, Pydantic, asyncio, MongoDB, PostgreSQL, Redis Streams, REST APIs
Cloud & DevOps	AWS, GCP, Kubernetes (EKS), Docker, Helm, Jenkins, GitHub Actions, CI/CD
Web & Practices	pytest, Unit Testing, React, Next.js, Supabase, Git, Agile/Scrum
Certifications	AWS Certified Solutions Architect – Associate, AWS Certified Cloud Practitioner

PROJECTS

codingplans: Rating LLM coding plans bought from Z.AI, MiniMax, and Kimi; tested KV caching, long context, and needle-in-a-haystack to show which providers nerfed caching or heavily quantized models.	
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